Computer Security Midterm Guidance

Week 1 Introduction to Computer Security

1. What do **confidentiality, integrity and availability** mean? Can you identify the main CIA goal affected in a short scenario?

2. Can you choose a suitable **control for each CIA goal**, such as access permissions, checks for changes, backups or redundancy?

3. What does **computer security** protect? Why can accidental errors and technical failures also cause security problems?

4. How do **people, process and technology** support security? Why use **defense in depth**, meaning several layers of protection?

5. What makes a security test **authorized**? What should you do when permission or the allowed scope is unclear?

Week 2 Assets Threats and Risk

1. Can you classify **assets** as information, devices, people or services? Can you explain their value and what they depend on?

2. Can you distinguish **threat, vulnerability, exploit, control, risk and incident**, and identify each one in a scenario?

3. Can you recognize **malicious, accidental and physical threats** and explain how each could harm an asset?

4. Can you rate **likelihood and impact** from 1 to 3, multiply the ratings, and use the class risk matrix to identify low, medium or high risk?

5. Why does a verified **patch** mainly reduce likelihood, while a successful **tested restore** mainly reduces impact?

Week 3 Social Engineering and Phishing

1. What is **social engineering**? Can you identify the invented story, requested action, and use of authority, fear, urgency, familiarity, reward or curiosity?

2. Can you distinguish **phishing, spear phishing, smishing, vishing and impersonation**? Why can one message fit more than one label?

3. Can you recognize **baiting, tailgating and deepfakes**? Why should a familiar face or voice still be checked before a sensitive action?

4. Can you spot warning signs in a message or full domain name, especially requests for **passwords, codes, money or remote access**?

5. Why do a familiar name, logo, fluent writing or **HTTPS** fail to prove that a request is genuine?

6. Can you apply **pause, verify and report** through an independent official route? Choose the response after sharing credentials, approving a login, installing an app or sending money.

Week 4 Identity Authentication and Access Control

1. Can you distinguish **identification, authentication and authorization** in a scenario? How do an account, a session and a permission differ?

2. How do **password guessing and credential stuffing** differ? Compare **online and offline guessing**, and know where login attempt limits help.

3. Why use **long, unique passwords** and a **password manager** instead of reusing passwords?

4. What are the three **MFA factor types**? Identify valid combinations and explain why password plus security question is not MFA.

5. Why must pages check **MFA completion**? What should you do with an unexpected login approval?

6. Can you apply **roles, scope and least privilege**? Distinguish **horizontal and vertical access failures**, and spot a permission violation in a log.

Week 5 Malware and Endpoint Security

1. Can you identify a **virus, worm, Trojan, ransomware, spyware or bot** from its behavior? How do a virus and a worm spread differently?

2. Why do symptoms alone not prove infection? Can malware stay hidden or combine several types?

3. Can you recognize infection paths through attachments, fake updates, unpatched services, USB devices or compromised suppliers?

4. Can you choose **updates, secure configuration, least privilege and endpoint protection** for a scenario? Why is antivirus alone insufficient?

5. Why must backups be **protected, separated and tested**? Why must recovery also fix the original weakness and return the system to a trusted state?

6. When you suspect infection, why **stop sensitive work, report and follow support guidance**?

Week 6 Networking Fundamentals for Security

1. What distinguishes **LAN, WAN and the Internet**? Can local printing work if Internet access fails?

2. Can you trace a request through a **client, access point, switch, router and server**? What does each device do, including the **default gateway**?

3. What do **IP addresses, ports, protocols, DNS and NAT** do? Distinguish private and public addresses, and recognize ports **443 for HTTPS** and **22 for SSH**.

4. Can you identify the scheme, domain and path in a **URL**? Explain the basic roles of **IP, TCP, HTTP and TLS**.

5. How does **HTTPS** protect data in transit? Why can phishing sites use it, and why does it not protect an infected endpoint?

6. Can you recognize an **evil twin, DNS spoofing, an on-path attacker or an unnecessarily exposed service**? Where does each risk occur?

7. Can you place suitable controls: verified Wi-Fi, **firewall rules, segmentation, valid HTTPS and server permissions**? What should you do at a certificate warning?

Instruction: Choose one best answer.

1. A person changes a student's recorded grade from 55 to 85 without permission. Which option identifies the **main CIA goal affected** and a suitable control?

A. Confidentiality: encrypt stored grades so unauthorized people cannot read them.

B. Availability: use a backup server to keep the grade system online.

C. Integrity: allow only authorized staff to edit grade records.

D. Confidentiality: use privacy screens to stop nearby people from seeing grades.

Correct answer: C.